

GRADE 7 SCIENCE — INTERACTIONS IN THE ENVIRONMENT

Ecosystems, Biotic & Abiotic Factors

Learning Goal

I can identify biotic and abiotic factors and explain how living and non-living parts of an ecosystem interact.

1. What Is an Ecosystem?

An **ecosystem** is a network of interactions between living organisms and their environment. An ecosystem includes **biotic factors** (living) and **abiotic factors** (non-living).

Examples: forest, pond, lake, grassland, ocean, desert, schoolyard.

2. Biotic and Abiotic Factors

Biotic = living: plants, animals, fungi, bacteria, insects

Abiotic = non-living: water, sunlight, air, temperature, soil, rocks, minerals, wind

A. Biotic or Abiotic?

Write **B** for biotic and **A** for abiotic.

1. Tree ____ 2. Sunlight ____ 3. Frog ____ 4. Water ____ 5. Soil ____
6. Mushroom ____ 7. Temperature ____ 8. Fish ____ 9. Rock ____ 10. Bacteria ____
11. Wind ____ 12. Grass ____ 13. Air ____ 14. Insect ____ 15. Minerals ____

B. Identify the Factors

Write **two biotic** and **two abiotic** factors you might find in a forest.

Biotic: _____, _____

Abiotic: _____, _____

C. Interactions

Living organisms interact with other organisms and with abiotic factors.

Complete each sentence.

1. Plants need _____ from the Sun to make food.
2. Fish need _____ to survive.
3. Plants absorb water and _____ from the soil.

4. A rabbit eats _____ to obtain energy.
5. Birds may use trees for _____ and shelter.
6. Temperature can affect which _____ can survive in an area.

D. Biotic–Abiotic Interactions

Explain how the abiotic factor affects the living organism.

1. A pond becomes very warm. What could happen to the fish?

2. A forest receives very little rain for several months. What could happen to plants?

3. A plant receives very little sunlight. How could this affect its growth?

4. A pond freezes during winter. How might this affect organisms in the pond?

E. Organism–Organism Interactions

Give an example of each interaction.

1. A bird eats an insect: _____
2. Two deer compete for food: _____
3. A bee gets nectar from a flower while helping pollinate it: _____
4. A fox hunts a rabbit: _____

What type of interaction is happening when one organism hunts and eats another?

F. Pond Ecosystem

Imagine a pond containing fish, frogs, algae, insects, bacteria, water, sunlight, temperature, rocks and oxygen.

1. Name three biotic factors: _____
2. Name three abiotic factors: _____
3. Which organism might eat insects? _____
4. Why is sunlight important to algae? _____
5. What could happen if the pond loses much of its water?

G. Cause and Effect

Complete the statements.

1. If there is less sunlight, plants may _____.
2. If plants decrease, animals that eat plants may _____.
3. If the number of insects increases, animals that eat insects may _____.
4. If a pond becomes polluted, organisms living in the pond may _____.

H. Multiple Choice

1. Which is an abiotic factor?
A. Frog B. Grass C. Temperature D. Bacteria
2. Which is a biotic factor?
A. Water B. Rock C. Sunlight D. Tree
3. Which best describes an ecosystem?
A. Only animals in an area
B. Only plants in an area
C. Living organisms and their interactions with the environment
D. Only non-living things
4. Which is an example of an organism–organism interaction?
A. Plant receiving sunlight
B. Fish swimming in water
C. Fox eating a rabbit
D. Plant absorbing minerals
5. Which change could affect many organisms in a pond?
A. A rock moves
B. Water becomes polluted
C. One leaf falls
D. One insect moves
6. Two rabbits need the same limited food source. This is an example of:
A. Competition B. Predation C. Decomposition D. Photosynthesis

I. True or False

1. ___ Biotic factors are living.
2. ___ Water is a biotic factor.
3. ___ Temperature is an abiotic factor.
4. ___ Organisms can interact with abiotic factors.
5. ___ An ecosystem contains only living organisms.
6. ___ A change in one part of an ecosystem can affect other parts.
7. ___ Animals can compete for limited resources.
8. ___ Sunlight can affect plant growth.

J. Challenge Question

A forest experiences a long period of drought.

1. Is drought related to a biotic or abiotic factor? _____
2. What could happen to the plants? _____
3. What could happen to animals that depend on those plants? _____

4. Explain how one change can affect several parts of an ecosystem.

K. Exit Ticket

Why are interactions important in an ecosystem?
